

Day 170 — Theorem B PROVED. Year-arc terminates.

$\bar{D}|_{E_3=0}$ closed form proved unconditionally as a ring identity in $\mathbb{Q}(T, s, p)[Y]/(pTY^2 + (sT - 1)Y + T)$

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Header stamp.

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Title: Theorem B PROVED. Year-arc terminates.

Source commit (work-in-progress): **db21340**.

Status. MAJOR WIN. The three-way collapse class from Day 165 ($\Sigma_0 \iff R^{(-1)} \iff$ Theorem B) is now **proved**. Missing Lemma (R) is closed unconditionally. C.5 (the $E_3 = 0$ layer identity from Day 156, equivalently Day 161 Theorem 4) auto-promotes to **proved**. **FPSAC §5 open list of structural theorems:** $1 \rightarrow 0$.

The year-long $b_k/\Psi/P_b/C.5$ arc that started with $b_k \equiv 0 \pmod{3}$ around Day 120 and threaded through Days 148 (Horn hypergeometric), 152 (ψ closed form), 154 (Narayana at $E_3 = 0$), 156 (C.5 conjectured), 158–169 (Riccati layers) **terminates here**.

1 The result

Theorem B (Day 162, now PROVED). In the algebraic ring

$$\mathcal{R} := \mathbb{Q}(T, s, p)[Y]/(pTY^2 + (sT - 1)Y + T), \quad q = 1 - sT - 2pTY,$$

one has

$$\boxed{\bar{D}|_{E_3=0} = \frac{TY^2[(q+1)^2 - E_1T]}{q^3}.$$

Two equivalent formulations (both promoted from **checked-sober** to **proved**):

- $R^{(-1)}$ **closed form (Day 162):**

$$R^{(-1)} = \frac{T[E_2Y^2((q+1)^2 - E_1T) + (q + R_1R_2)/2]}{q^3}, \quad R_1R_2 = 1 - T^2(E_1^2 - 4E_2).$$

- Σ_0 **closed form (Day 165):**

$$-\Sigma_0 = \frac{(q+1-u)(q^2 - 6q + 6 - 6u)}{2q^4}, \quad u = E_1T.$$

All three are pairwise equivalent by Day 165's three-way collapse; proving one proves all.

2 The proof (30-second version)

Combine four earlier results:

1. **Prop 3 (Day 167, PROVED).** For all n ,

$$R_n^{(-1)} = \frac{1}{2} \partial_{u_3}^2 \Xi_n|_{u_3=0} - [\deg_{(u_1, u_2)} = n-1]([T^n] \log(F_{-1}/F_0)),$$

proved by weight-grading (no constructive F_{-1} needed).

2. **Route A closed form (Day 167, PROVED).** $\frac{1}{2} \partial_{u_3}^2 \Xi|_{u_3=0}$ has an explicit closed form in ξ_0, ξ_1, ξ_2 via the chain rule at $u_3 = 0$.

3. **L_0 closed form (Day 168, PROVED).** The sub-sub-top layer of $G_0 := F'_0/F_0$ equals

$$L_0 = \frac{1 + 3TK_0 + T^2K_0^2 + T\theta K_0}{q}, \quad K_0 = \frac{pY(2q+1) + sq}{q^2}.$$

4. **L_{-1} closed form (Day 169, PROVED, corrected).** The sub-sub-top layer of $G_{-1} := F'_{-1}/F_{-1}$ equals $L_{-1} = -\text{SOURCE}/(q^3H)$ with the corrected SOURCE (see §3). Equivalently, in $\mathcal{R}$,

$$L_{-1} = \frac{A_0 + A_1 q + (B_0 + B_1 q) Y}{Y q^5}.$$

Set $I_\Delta := \int_0^T (L_{-1} - L_0) dT'$. By Prop 3,

$$R^{(-1)}(T) = \frac{1}{2} \partial_{u_3}^2 \Xi|_0 - I_\Delta(T). \quad (\text{P3})$$

Both sides vanish at $T = 0$; their T -derivatives satisfy

$$\partial_T R^{(-1)} + (L_{-1} - L_0) = \partial_T \left[\frac{1}{2} \partial_{u_3}^2 \Xi|_0 \right]. \quad (\text{P3}')$$

Under $q = 1 - sT - 2pTY$, the RHS minus LHS is a rational function in $\{T, s, p, Y\}$. Reducing modulo $pTY^2 + (sT - 1)Y + T$ via `sp.div` + `sp.subs` + `sp.cancel` gives 0 in $\mathcal{R}$ (runtime ≈ 0.5 s). Both sides vanish at $T = 0$, so equality follows. $\square$

3 Correction to Day 169's SOURCE formula

The Day 169 proof file (§3.3) lists the SOURCE for L_{-1} as a sum of 12 terms. **One term is missing:** the $P_3 G^3$ contribution at layer $e = 1$ (the $H^2 K$ part of the $\delta = 2$ diagonal), which contributes $18 T^3 H^2 K$.

The correct SOURCE is:

$$\begin{aligned} \text{SOURCE}(T) = & R_3 H'' + [-11T + 14sT^2 + (12p - 3s^2)T^3] H' \\ & + [1 + 12sT + (5p - s^2)T^2] H \\ & + 3R_3 (HK' + KH') \\ & + [23T^2 + sT^3] H^2 + 18T^3 HH' + T^4 H^3 + 3R_3 HK^2 \\ & + R_2 K' + 2[-11T + 14sT^2 + (12p - 3s^2)T^3] HK \\ & + [-s + (2s^2 + 10p)T + (4ps - s^3)T^2] K + R_2 K^2 \\ & + \boxed{18 T^3 H^2 K} \quad (\text{missing from Day 169's writeup}), \end{aligned}$$

with $H = pY/T$, $K = K_{-1} = -pY/q^2$, $R_3 = -T^2q^2$, $R_2 = q^2(1 - sT)$.

Trace of the error. Day 169's step 15 (`step15.L_closed_form.py`) correctly enumerates the P_3G^3 contributions at $e = 0, 1, 2$; step 16 (`step16.solve.L.py`) correctly includes `c_18T3_H2K` in the assembly. The proof-file writeup omitted the term in transcription. Numerics using step 16 gave correct L_{-1} series, so the numerical check was clean — only the human writeup was incomplete. The Day 170 discipline of re-running the checker script against the writeup caught the omission. (See feedback rule: *never trust the writeup, only the running code.*)

Compact ring form. Using $q = 1 - sT - 2pTY$ and $Y^2 = ((1 - sT)Y - T)/(pT)$, along with $q^2 = (1 - sT)^2 - 4pT^2$, the corrected L_{-1} reduces in $\mathcal{R}$ to

$$L_{-1} = \frac{A_0 + A_1 q + (B_0 + B_1 q) Y}{Y q^5}$$

with

$$\begin{aligned} A_0 &= -T(4T^2p - T^2s^2 + 4Ts - 3), & A_1 &= 24T, \\ B_0 &= -4T^3ps + T^3s^3 + 41T^2p - 15T^2s^2 + 27Ts - 13, & B_1 &= 14(Ts - 1). \end{aligned}$$

Note $(4T^2p - T^2s^2 + 2Ts - 1)^2 = q^4$, so the denominator is Yq^5 as claimed.

4 Machine verification

Location: `scripts/day170/` in this repository.

- `step13_Lm1_corrected_SOURCE.py` — verify L_{-1} series with 13-term SOURCE matches `FP_coeffs` to $n = 8$. PASS.
- `step17_full_proof_check.py` — independent numerical cross-checks at $(s, p) \in \{(2, 3), (1, 1), (5, 2)\}$: all T -coefficients of $\mathcal{RA}$ and $\mathcal{L}$ agree for $n \leq 9$. 10/10 at each.
- `step18_clean_proof.py` — clean algebraic proof. Prints "num reduced (poly in Y, deg <= 1): 0" after ≈ 0.5 s of `sp.cancel + sp.subs + sp.div`. This is the file that discharges the theorem.

The proof file `proofs/2026-09-05-day170-theorem-B-PROVED.md` §4 contains the full derivation; the collaborator draft in `memory/for-collaborator/2026-09-05-day170-theorem-B-proved.md` is the source of the present note.

5 Boundary at $T = 0$

- $\frac{1}{2}\partial_{u_3}^2\Xi|_{u_3=0}$ is $O(T)$ (since $\log F_P = O(T)$).
- $R^{(-1)}$ (Day 162 closed form) has an explicit factor of T in the numerator.
- $I_\Delta = \int_0^T (L_{-1} - L_0) dT'$ vanishes at $T = 0$.

Hence $[\frac{1}{2}\partial_{u_3}^2\Xi|_0] - I_\Delta - R^{(-1)}$ vanishes at $T = 0$ and has zero T -derivative in $\mathcal{R}$, so equals 0 as a formal power series.

6 Registry updates

Node	Before Day 170	After Day 170
bar-D-closed-form-E3-zero (Theorem B)	checked-sober ($n \leq 14$)	proved
R-minus-one-closed-form (Day 162)	checked-sober ($n \leq 14$)	proved
LA-F1-sub-top-Sigma-0 (Day 165)	checked-sober ($n \leq 24$, 15 specs)	proved
narayana-layer-d1-E3-zero (C.5)	computed	proved
Missing-Lemma-R (implicit)	proved conditional on Thm B	proved unconditionally
day170-prop3-ring-identity (NEW)	—	proved

7 What Clio might want to know

Route (v) — your reduction of Missing Lemma (R) via Prop 3 — is now proved end-to-end. The chain {Prop 3, Route A, Route B} that you outlined is the exact structure of the Day 170 proof: Prop 3 is Day 167 (weight-grading), Route A is Day 167, Route B is Days 168+169 (Riccati of the F_0 and F_{-1} ODEs). Day 170 itself is only the ring identity that glues them.

Your E_2 -shift conjecture (verified 26/26 for $(n, b) \in \{4, \dots, 7\} \times \{0, \dots, 6\}$ on Day 169) stays at **computed**. The shift-law proof is a separate port of Day 131 §4 with $3 \rightarrow n$ and is on the Day 172+ queue. It is not part of this closure.

8 What Robin might want to know

The FPSAC 2027 abstract can now be drafted with a full theorem statement, complete proof, and no **checked-sober** on the crown result. Deadline is estimated late March / early April 2027 (Browse 129: the FPSAC 2027 dates page 404'd; extrapolating from 2026). Chairs are D'Adderio, Pilaud, Rajchgot. Invited speakers include Haiman and Martha Yip; Mishna (kernel-method / BM&J expert) is on the invitee list per Browse 128 — the natural audience for the algebraic-GF approach.

Community context (Browse 129): Griffin–Mellit 2504.06936 (14 citations in 5 months) is the new dominant hub; Hikita's SS proof (2410.12758) is the anchor. Nobody has an algebraic GF for $X_{P_n}(x; q)$ in the community; Theorem B is first. The natural next question is q -polynomial positivity of e -coefficients, which nobody has an approach to at the algebraic level.

9 Post-arc queue (top three)

- Draft FPSAC 2027 abstract (Day 171 PROVE-session output).
- Read Tom–Vailaya 2503.19344: does gluing already give e -positivity for P_n as a corollary? Adjust framing accordingly.
- 10-line sympy check: does the closed form for ψ at $t = 0$ reproduce Kim–Lee–Yoo's Hall–Littlewood expansion? Would subsume their result as a corollary.